MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS, FALL SEMESTER 2025
CHAPTER 3: LIMITS AND CONTINUITY (PART 2)

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Goals of this section:

(1) Continuous functions
(2) Construction of functions
(3) Examples (Showing some known functions to be continuous)
(4) Continuous functions on an interval: boundedness, maximum-minimum theorem, preserving intervals
(5) Uniform Continuity

1. Continuous functions

1.1. Definition.

Definition 1.1. (1) Let $A \subset \mathbb{R}$, let $f : A \to \mathbb{R}$. Let $c \in A$. We say that f is continuous at c if for any given $\epsilon > 0$, there exists $\delta > 0$ such that for any $x \in A$ with $|x - c| < \delta$,

$$|f(x) - f(c)| < \epsilon.$$

(2) If f is not continuous at c, we also say that f is discontinuous at c.
(3) For a subset $B \subset A$, we say that f is continuous on B if f is continuous at every point of B.

Remark 1.2. When $c \in A$ is a cluster point of A, the above definition is equivalent to requiring that

$$\lim_{x \to c} f(x)$$

exists and equals $f(c)$. If $c \in A$ is not a cluster point of A, then c is called an isolated point of A. Isolated points are automatically points of continuity; see the following exercise.

Exercise 1.3. Let $f : A \to \mathbb{R}$. Let $c \in A$ be an isolated point in A. Prove that f is continuous at c by $\epsilon - \delta$ definition.

Using Remark 1.2 and Exercise 1.3, we have:

Exercise 1.4. Let $f : A \to \mathbb{R}$. Let $c \in A$. Then f is continuous at c if and only if either

(1) c is a cluster point of A and $\lim_{x\to c} f(x) = f(c)$;
(2) c is an isolated point of A.

Exercise 1.5. (Sequential criterion) Let $f : A \to \mathbb{R}$, let $c \in A$. Then f is continuous at c if and only if for every sequence (x_n) in A that converges to c, the sequence $(f(x_n))$ converges to $f(c)$.

Exercise 1.6. Let $f : A \to \mathbb{R}$, let $c \in A$. Then f is discontinuous at c if and only if there exists a sequence (x_n) in A such that (x_n) converges to c and the sequence $(f(x_n))$ does not converge to $f(c)$.

1.2. **Examples.** Several typical examples can be shown to be continuous by the previous discussions on limits.

Example 1.7.
(1) (Constant function) For $b \in \mathbb{R}$, let $f(x) = b$. Then, for any $c \in \mathbb{R}$, $\lim_{x\to c} f = b = f(c)$. Hence, f is continuous at any c.
(2) Let $f(x) = x$. For any $c \in \mathbb{R}$, $\lim_{x\to c} f(x) = c = f(c)$. Hence, f is continuous at every point in $\mathbb{R}$.

Example 1.8. (Dirichlet discontinuous function) Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is rational,} \\ 0, & \text{if } x \text{ is irrational.} \end{cases}$$

Then f is discontinuous at every point of $\mathbb{R}$.
Proof: Let $c \in \mathbb{R}$. We consider two cases.

(a) Suppose c is rational. For each $n \in \mathbb{N}$, let
$$x_n = c + \frac{\sqrt{2}}{n}.$$
Then x_n is irrational and $x_n \to c$. Hence $f(x_n) = 0$ for all n, while $f(c) = 1$. Thus $(f(x_n))$ does not converge to $f(c)$.
(b) Suppose c is irrational. By the density of $\mathbb{Q}$ in $\mathbb{R}$, there exists a sequence (x_n) of rational numbers such that $x_n \to c$. Then $f(x_n) = 1$ for all n, while $f(c) = 0$. Thus $(f(x_n))$ does not converge to $f(c)$.

By Exercise 1.6, f is discontinuous at c.

Exercise 1.9. For $x \in \mathbb{R}$, define $[[x]]$ to be the greatest integer n such that $n \le x$. Determine the points of continuity of the following functions:

(1) $f(x) = [[x]]$
(2) $f(x) = [[\sin(x)]]$

1.3. **Extending a continuous function.** Let $f : A \to \mathbb{R}$ be continuous. Let c be a cluster point of A. Suppose $c \notin A$. If $\lim_{x\to c} f$ exists, then f can be extended to a continuous function $\widetilde{f} : A \cup \{c\} \to \mathbb{R}$ such that:

$$\widetilde{f}(x) = \begin{cases} f(x) & x \in A \\ \lim_{x\to c} f & x = c \end{cases}$$

Example 1.10. Let

$$f(x) = x \sin\left(\frac{1}{x}\right)$$

defined for $x \neq 0$. For $x \neq 0$, since

$$-|x| \leq x \sin\left(\frac{1}{x}\right) \leq |x|,$$

the squeeze theorem gives that

$$\lim_{x\to 0} x \sin\left(\frac{1}{x}\right) = 0.$$

We can extend f to the function

$$\widetilde{f}(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

which is continuous at 0.

Exercise 1.11. Let $f(x) = \frac{x^2+x-12}{x-3}$ $(x \neq 3)$. Extend f to a function defined also at $x = 3$.

2. Construction of continuous functions

2.1. **Basic operations.**

Theorem 2.1. *Let $c \in A$. Let $f, g : A \to \mathbb{R}$ be continuous functions at c. Let $b \in \mathbb{R}$. Then*

1. *$f + g$, $f - g$, bf and fg are continuous functions at c; and*
2. *if $g(c) \neq 0$, then there exists $\delta > 0$ such that $g(x) \neq 0$ for all $x \in A \cap (c - \delta, c + \delta)$, and $\frac{f}{g}$ is continuous at c.*

Proof. As shown before, if c is an isolated point, then there is nothing to prove. Suppose c is a cluster point.

$$\lim_{x\to c}(f \pm g) = \lim_{x\to c} f \pm \lim_{x\to c} g = f(c) \pm g(c) = (f \pm g)(c),$$

$$\lim_{x\to c}(fg) = (\lim_{x\to c} f)(\lim_{x\to c} g) = f(c)g(c) = (fg)(c).$$

Since g is continuous at c and $g(c) \neq 0$, there exists $\delta > 0$ such that

$$|g(x) - g(c)| < \frac{|g(c)|}{2}$$

whenever $x \in A$ and $|x - c| < \delta$. Hence $g(x) \neq 0$ for all such x. Since we also have $g(c) \neq 0$,

$$\lim_{x \to c} \left(\frac{f}{g} \right) = \frac{\lim_{x \to c} f}{\lim_{x \to c} g} = \frac{f(c)}{g(c)}.$$

□

Exercise 2.2. Deduce bf is a continuous function from that fg is a continuous function.

2.2. **Composition of functions.** For a function $f : A \to \mathbb{R}$ and a function $g : B \to \mathbb{R}$, suppose $f(A) \subset B$ and define

$$(g \circ f)(x) = g(f(x)).$$

Theorem 2.3. *Let $f : A \to \mathbb{R}$ and let $g : B \to \mathbb{R}$. Suppose $f(A) \subset B$. If f is continuous at $c \in A$ and g is continuous at $f(c)$, then $g \circ f$ is continuous at c.*

Proof. Let (x_n) be any sequence in A converging to c. By the sequential criterion, $f(x_n)$ converges to $f(c)$. Since g is continuous at $f(c)$, we have that $g(f(x_n))$ converges to $g(f(c))$. Thus $(g \circ f)(x_n)$ converges to $(g \circ f)(c)$. By the sequential criterion, $g \circ f$ is continuous at c. □

Exercise 2.4. Prove Theorem 2.3 by using the ϵ-δ definition of continuity.

3. Examples of continuous functions

Example 3.1.

(a) Any polynomial function is continuous. This follows from scalar multiplication, addition, multiplication, and composition, starting from the continuous function $f(x) = x$. (Example 1.7).

(b) Let $p(x)$ and $q(x)$ be polynomials. Suppose $q(c) \neq 0$. Let $f(x) = \frac{p(x)}{q(x)}$. It is shown before that, if $q(c) \neq 0$,

$$\lim_{x \to c} \frac{p(x)}{q(x)} = \frac{\lim_{x \to c} p(x)}{\lim_{x \to c} q(x)} = \frac{p(c)}{q(c)}.$$

Hence, f is continuous at c.

(c) Let $f(x) = \sin(x)$. We shall use the following two algebraic properties:

$$|\sin(x)| \leq |x|,$$

$$|\cos(x)| \leq 1,$$

and

$$\sin(x) - \sin(y) = 2 \sin\left(\frac{1}{2}(x - y) \right) \cos\left(\frac{1}{2}(x + y) \right).$$

Then,

$$\begin{aligned} |\sin(x) - \sin(c)| &= 2|\sin\left(\frac{1}{2}(x-c)\right)\cos\left(\frac{1}{2}(x+c)\right)| \\ &\leq |x-c| \end{aligned}$$

Since $\lim_{x\to c}|x-c| = 0$, the squeeze theorem implies that

$$\lim_{x\to c}|\sin(x) - \sin(c)| = 0.$$

and so

$$\lim_{x\to c}\sin(x) = \sin(c).$$

Thus, sin is continuous on $\mathbb{R}$.

Remark. As an extension of Example 3.1, for $n, m \in \mathbb{N}$, $f(x) = x^{m/n}$ is continuous on its natural domain. A rigorous proof requires more than what we discussed so far. But one may freely use this later.

Exercise 3.2. Let $f(x) = \sqrt{x+\sqrt{x}}$ for $x \geq 0$. Use Theorems 2.1 and 2.3 to explain that f is continuous on $[0, \infty)$.

Example 3.3. (1) Show that $f(x) = |x|$ is continuous.

(2) Let $f : A \to \mathbb{R}$ be continuous. Define $|f| : A \to \mathbb{R}$ be given by $|f|(x) = |f(x)|$ for all $x \in A$. Show that $|f|$ is continuous.

Exercise 3.4. Let f and g be continuous functions. Let $h(x) = \sup\{f(x), g(x)\}$. Then h is a continuous function. (Hint: use $h(x) = \frac{f(x)+g(x)}{2} + \frac{|f(x)-g(x)|}{2}$.)

Exercise 3.5. (Composition of discontinuous function and a continuous function) Give an example of $f : \mathbb{R} \to \mathbb{R}$ which is discontinuous at every point and $|f|$ is continuous at every point.

4. Boundedness of continuous functions

Definition 4.1. A function $f : A \to \mathbb{R}$ is said to be bounded on A if there exists a number $M > 0$ such that

$$|f(x)| < M$$

for all $x \in A$. If no such M exists, then f is said to be unbounded on A.

Exercise 4.2. Let f be an unbounded function. Show that there exists a sequence (x_n) in A such that $|f(x_n)| > n$.

We now prove that every continuous function on a closed bounded interval is bounded.

Theorem 4.3. *(Boundedness of a continuous function on a closed bounded interval) Let I be a closed bounded interval. Let $f : I \to \mathbb{R}$ be a continuous function. Then f is bounded on I.*

Proof. We prove this by contradiction. Suppose f is unbounded.

Step 1: Obtain an unbounded sequence

By Exercise 4.2, we can find a sequence (x_n) in I such that

$$(*) \quad |f(x_n)| > n.$$

Step 2: Obtain a convergent subsequence

Since I is bounded, (x_n) is bounded. By the Bolzano-Weierstrass Theorem, there exists a convergent subsequence (x_{n_k}) of (x_n). Since I is closed and $x_{n_k} \in I$ for all k, its limit belongs to I. Let $x_{n_k} \to x^* \in I$. Since f is continuous at x^*, the sequence $(f(x_{n_k}))$ converges, and hence is bounded.

Hence, $|f(x_{n_k})| < M$ for a number $M > 0$.

Step 3: Obtain a contradiction

By $(*)$, we have

$$|f(x_{n_k})| > n_k$$

for all k. Since $n_k \to \infty$, the sequence $(f(x_{n_k}))$ is unbounded. This contradicts Step 2.

□

Exercise 4.4. Find an example that $I = [a, \infty)$ and $f : I \to \mathbb{R}$ is a continuous function, but f is not bounded on I.

Exercise 4.5. Find an example that $I = (a, b]$ for some $a, b \in \mathbb{R}$, and $f : I \to \mathbb{R}$ is a continuous function, but f is not bounded on I.

Exercise 4.6. Find an example that I is a closed interval and $f : I \to \mathbb{R}$ is a continuous function, but f is not bounded on I.

5. The maximum-minimum theorem (a.k.a. Extreme Value Theorem)

5.1. The maximum-minimum theorem.

Definition 5.1. Let $f : A \to \mathbb{R}$. We say that f has an absolute maximum on A if there exists $x^* \in A$ such that for all $x \in A$,

$$f(x^*) \geq f(x).$$

We say that f has an absolute minimum on A if there exists $x_* \in A$ such that for all $x \in A$,

$$f(x_*) \leq f(x).$$

Example 5.2. (1) Let $f : [-\frac{1}{2}, 1] \to \mathbb{R}$ be given by $f(x) = x^2$. Then f has the absolute maximum at $x = 1$ and f attains the absolute minimum at $x = 0$.

(2) Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = \sin(x)$. Then f has the absolute maximum at points $x = 2n\pi + \frac{\pi}{2}$ $(n \in \mathbb{Z})$ and the absolute minimum at points $x = 2n\pi - \frac{\pi}{2}$ $(n \in \mathbb{Z})$.

In general, a continuous function may not have an absolute maximum/minimum:

Exercise 5.3. (1) Give an example of a continuous bounded function $f : (0, 1) \to \mathbb{R}$ which does not attain an absolute maximum.

(2) Give an example of a continuous function $f : [0, \infty) \to \mathbb{R}$ which does not attain an absolute minimum.

Theorem 5.4. *(Maximum-minimum theorem) Let I be a closed bounded interval, and let $f : I \to \mathbb{R}$ be continuous. Then f attains an absolute maximum and an absolute minimum on I.*

Proof. We prove only the statement for absolute maximum, and a proof for the statement for absolute minimum is similar.

Step 1: Use $\sup f(I)$ to construct a sequence (x_n)

By Theorem 4.3, $f(I)$ is bounded. Then $\sup f(I)$ exists and let $s^* = \sup f(I)$. For any $n > 0$, there exists $x_n \in I$ such that $s^* - \frac{1}{n} < f(x_n) \leq s^*$.

Step 2: Use the Bolzano-Weierstrass Theorem to construct a convergent sequence

Now, by the Bolzano-Weierstrass Theorem, there exists a convergent subsequence (x_{n_k}) of (x_n). Let $x^* = \lim_{k\to\infty} x_{n_k}$.

Step 3: Show $x^* \in I$

Moreover, let $I = [a, b]$ and then we have $a \leq x_{n_k} \leq b$ for all $k \in \mathbb{N}$, and so it is shown in Chapter 2 that

$$a \leq x^* \leq b$$

and so $x^* \in I$.

Step 4: Show $f(x^*) = s^*$

We have the inequality $s^* - \frac{1}{n_k} < f(x_{n_k}) \leq s^*$ for all k. Since $\lim_{k\to\infty}(s^* - \frac{1}{n_k}) = s^*$, the squeeze theorem gives that

$$\lim_{k\to\infty} f(x_{n_k}) = s^*$$

By the limit criterion of continuity (Exercise 1.5), we have $f(x^*) = s^*$. □

Exercise 5.5. Let $I = [a, b]$, and let $f : I \to \mathbb{R}$ be a continuous function such that $f(x) > 0$ for all $x \in I$. Prove that there exists $m > 0$ such that $f(x) \geq m$ for all $x \in I$.

5.2. **Bolzano intermediate value theorem.** We now move to some intermediate values between the absolute maximum and the absolute minimum. One may also rephrase to the question to find the roots of a function f i.e. solve $f(x) = 0$. The following theorem and its proof indicate a method, called the bisection method. Application to a continuous function on an interval will be given in Corollary 5.11.

Theorem 5.6. *Let $I = [a, b]$ and let $f : I \to \mathbb{R}$ be a continuous function. If either $f(a) < 0 < f(b)$ or $f(a) > 0 > f(b)$, then there exists $c \in I$ such that $f(c) = 0$.*

Proof. Step 1: Construct sequences by bisections
We only consider the case that $f(a) < 0 < f(b)$. We prove this by contradiction. Suppose $f(c) \neq 0$ for all $c \in I$. Let $a_1 = a$ and let $b_1 = b$. Then, we have either $f(\frac{a_1+b_1}{2}) > 0$ or $f(\frac{a_1+b_1}{2}) < 0$.

(1) If $f(\frac{a_1+b_1}{2}) > 0$, set $a_2 = a$ and $b_2 = \frac{a_1+b_1}{2}$.
(2) If $f(\frac{a_1+b_1}{2}) < 0$, set $a_2 = \frac{a_1+b_1}{2}$ and $b_2 = b$.

In either case, we have $f(a_2) < 0 < f(b_2)$.

Similarly, we can find sequences (a_n) and (b_n) such that for all $n \in \mathbb{N}$,

$$f(a_n) < 0 < f(b_n)$$

with the following properties:

(1) $a_n \leq a_{n+1}$ and $b_{n+1} \leq b_n$
(2) $b_n - a_n = \frac{1}{2^{n-1}}(b_1 - a_1)$.

Step 2: Show the limits (a_n) and (b_n) exist
Since (a_n) is increasing and is bounded above, the MCT implies that $\lim\ a_n$ exists. Since (b_n) is decreasing and is bounded below, the MCT implies that $\lim\ b_n$ exists. The second condition implies that $\lim\ (b_n - a_n) = 0$ and so $\lim\ b_n = \lim\ a_n$.

Step 3: Find limits of $f(a_n)$ and $f(b_n)$ Let $c = \lim\ a_n = \lim\ b_n$. Since f is continuous, $\lim\ f(a_n) = \lim\ f(b_n) = f(c)$. Since $f(a_n) < 0$ for all n, we have

$$\lim_{n\to\infty} f(a_n) \leq 0.$$

Since $f(b_n) > 0$ for all n, we have

$$\lim_{n\to\infty} f(b_n) \geq 0.$$

But $a_n \to c$ and $b_n \to c$, so by continuity,

$$\lim_{n\to\infty} f(a_n) = f(c) = \lim_{n\to\infty} f(b_n).$$

Therefore $f(c) \leq 0$ and $f(c) \geq 0$, hence $f(c) = 0$, a contradiction. □

Theorem 5.7. *(Bolzano's intermediate value theorem) Let I be an interval and let $f : I \to \mathbb{R}$ be continuous. Let $a, b \in I$ with $f(a) < f(b)$, and let $k \in (f(a), f(b))$. Then there exists a point c between a and b such that $f(c) = k$.*

Proof. Let $g(x) = f(x) - k$. Assume first that $a < b$. The case $b < a$ is similar. Then $g(a) < 0$ and $g(b) > 0$. Since g is still continuous, by Theorem 5.6, there exists c such that $g(c) = 0$. Hence, $f(c) = k$. □

Example 5.8. Let $f(x) = x^6 - 10x + 1$. Show that f has at least two real roots.
Sol: Note that $f(100) > 0$ and $f(-100) > 0$. We also have $f(1) = -8$. By the Bolzano intermediate theorem, there is $c_1 \in (1, 100)$ such that $f(c_1) = 0$ and there is $c_2 \in (-100, 1)$ such that $f(c_2) = 0$.

Exercise 5.9. Use Bolzano's intermediate value theorem to show that every polynomial of odd degree with real coefficients has at least one root.

Exercise 5.10. Let $f(x) = (x-1)(x-2)(x-3)(x-4)(x-5)$. Use the bisection method on the interval $[0, 7]$. Which roots will be located by the bisection method?

5.3. **Continuous functions preserving intervals.**

Corollary 5.11. *Let I be an interval. Let $f : I \to \mathbb{R}$ be a continuous function. Then, $f(I)$ is also an interval.*

Proof. Let $\alpha, \beta \in f(I)$ with $\alpha < \beta$. Then, $f(a) = \alpha$ and $f(b) = \beta$ for some $a, b \in I$. Now, for any $\gamma \in [\alpha, \beta]$, Bolzano's intermediate value theorem implies that $f(c) = \gamma$ for some $c \in I$. This implies that $[\alpha, \beta] \subset f(I)$. This implies that $f(I)$ is an interval □

Example 5.12. Let $f(x) = \sin(x)$. Then $f(\mathbb{R}) = [-1, 1]$, $f([0, \pi]) = [0, 1]$, $f((0, \pi)) = (0, 1]$.

Exercise 5.13. Let I be an interval. Let $f : I \to \mathbb{R}$ be a continuous function. Determine if the following statements are true:

(1) Let I be a closed bounded interval. Then $f(I)$ is also a closed bounded interval.
(2) Let I be an open bounded interval. Then $f(I)$ is an open bounded interval.

6. Uniform continuity

6.1. **Definitions.**

Definition 6.1. Let $f : A \to \mathbb{R}$. We say that f is uniformly continuous if for any given $\epsilon > 0$, there exists $\delta > 0$ such that for any $x, y \in A$, if $|x - y| < \delta$,

$$|f(x) - f(y)| < \epsilon.$$

Exercise 6.2. Show that a uniformly continuous function $f : A \to \mathbb{R}$ is continuous on A.

Exercise 6.3. Compare with the definition of a continuous function. Do you know why the term 'uniform' is used?

Example 6.4.

(1) Define $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = x$. Then f is uniformly continuous. To see this, for any given $\epsilon > 0$, we choose $\delta = \epsilon$ such that for any $|x - y| < \delta = \epsilon$, $|f(x) - f(y)| = |x - y| < \epsilon$.

(2) Define $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = x^2$. Then f is not uniformly continuous. We choose $\epsilon = 1$. For any $\delta > 0$, we choose $x = \frac{1}{\delta}$ and $y = \frac{1}{\delta} + \frac{\delta}{2}$. Then $|x - y| < \delta$.
But $|f(x) - f(y)| = |x - y||x + y| = \frac{\delta}{2}\left(\frac{2}{\delta} + \frac{\delta}{2}\right) = 1 + \frac{\delta^2}{4} > 1$.

(3) Define

$$f(x) = \begin{cases} 1 & x > 0 \\ -1 & x \leq 0 \end{cases}$$

Since f is not continuous at 0, f is not uniformly continuous.

Exercise 6.5. Draw a diagram to understand why example (2) above fails to be uniformly continuous.

Exercise 6.6. Let $f : A \to \mathbb{R}$ and let $g : A \to \mathbb{R}$ be uniformly continuous. Show that $f + g$ is also uniformly continuous.

In general, for two uniformly continuous functions $f, g : A \to \mathbb{R}$, fg is not necessarily uniformly continuous.

6.2. **Non-uniform continuity criterion.** As before, one may formulate some sequential criterion, which is useful to show non-uniform continuity:

Exercise 6.7. (Non-uniform Continuity) Let $f : A \to \mathbb{R}$. Then the following statements are equivalent:

(1) f is not uniformly continuous on A.
(2) There exists $\epsilon_0 > 0$ such that for any $\delta > 0$, there exist $x_\delta, y_\delta \in A$ such that $|x_\delta - y_\delta| < \delta$ and $|f(x_\delta) - f(y_\delta)| \geq \epsilon_0$.
(3) There exists $\epsilon_0 > 0$ such that there exist two sequences (x_n) and (y_n) of numbers in A such that $\lim(x_n - y_n) = 0$, and for all $n \in \mathbb{N}$, $|f(x_n) - f(y_n)| \geq \epsilon_0$.

6.3. **Uniform continuous theorem.** We shall introduce a family of uniformly continuous functions:

Theorem 6.8. *(Closed+bounded+continuous ⇒ Uniformly continuous) Let I be a closed bounded interval. If $f : I \to \mathbb{R}$ is continuous, then f is uniformly continuous.*

Proof. We prove this by contradiction. Suppose f is not uniformly continuous.
Step 1: Construct two sequences
By Exercise 6.7, there exist two sequences (x_n) and (y_n) such that $\lim(x_n - y_n) = 0$ and $|f(x_n) - f(y_n)| \geq \epsilon_0$ for some $\epsilon_0 > 0$.

Step 2: Construct convergent subsequences
By the Bolzano-Weierstrass Theorem, there exists a subsequence (x_{n_k}) such that $x_{n_k} \to z$ for some $z \in I$. Since

$$|x_{n_k} - y_{n_k}| \to 0,$$

we also have $y_{n_k} \to z$. Since I is closed, $z \in I$.
Step 3: Use continuity
Since f is continuous at z, we have

$$f(x_{n_k}) \to f(z) \quad \text{and} \quad f(y_{n_k}) \to f(z).$$

Hence

$$|f(x_{n_k}) - f(y_{n_k})| \to 0,$$

which contradicts the fact that

$$|f(x_{n_k}) - f(y_{n_k})| \geq \epsilon_0$$

for all k.

□

Example 6.9. The following functions are uniformly continuous:

(a) $f : [1, 2] \to \mathbb{R}$, $f(x) = x^2$
(b) $f : [0, 100] \to \mathbb{R}$, $f(x) = \frac{\sin(x)}{x^2+1}$
(c) $f : [2, 10] \to \mathbb{R}$, $f(x) = \sin\left(\frac{1}{x}\right)$.

Exercise 6.10. Show that the function $f : [0, \infty) \to \mathbb{R}$ be given by

$$f(x) = \frac{1}{1 + x^2}$$

is uniformly continuous.

Exercise 6.11. A function $f : \mathbb{R} \to \mathbb{R}$ is said to be periodic if there exists $T > 0$ such that $f(x + T) = f(x)$ for any $x \in \mathbb{R}$. Show that any continuous periodic function is uniformly continuous.

6.4. **Lipschitz functions.** We now describe another family of uniformly continuous functions.

Definition 6.12. Let $f : A \to \mathbb{R}$ be a function. We say that f is a Lipschitz function if there exists a constant $K > 0$ such that for any $x, y \in A$,

$$|f(x) - f(y)| \leq K|x - y|.$$

Exercise 6.13. Understand Definition 6.12 geometrically by interpreting the value $\frac{f(x)-f(y)}{x-y}$ as a slope between points $(x, f(x))$ and $(y, f(y))$.

Example 6.14. Let $f : [1, \infty) \to \mathbb{R}$ be given by $f(x) = \sqrt{x}$. Now, for any $x, y \geq 1$,

$$|f(x) - f(y)| = |\sqrt{x} - \sqrt{y}| = \frac{|x - y|}{\sqrt{x} + \sqrt{y}} \leq \frac{1}{2}|x - y|,$$

where the last inequality follows from

$$\frac{1}{\sqrt{x} + \sqrt{y}} \leq \frac{1}{2}.$$

Hence, f is Lipschitz.

Example 6.15. Let $g : [0, 2] \to \mathbb{R}$ be given by $g(x) = \sqrt{x}$. If g were Lipschitz, then there would exist $K > 0$ such that

$$|\sqrt{x} - \sqrt{y}| \leq K|x - y|$$

for all $x, y \in [0, 2]$. Taking $x = 0$ and $y = t > 0$, we get

$$\sqrt{t} \leq Kt,$$

so

$$\frac{1}{\sqrt{t}} \leq K.$$

Letting $t \to 0^+$ gives a contradiction. Hence g is not Lipschitz.

Theorem 6.16. *Let $f : A \to \mathbb{R}$ be a Lipschitz function. Then f is uniformly continuous.*

Proof. Since f is Lipschitz, there exists a constant $K > 0$ such that for all $x, y \in A$,

$$|f(x) - f(y)| \leq K|x - y|.$$

Let $\epsilon > 0$ be given. Choose $\delta = \epsilon/K$. Then, whenever $x, y \in A$ and $|x - y| < \delta$, we have

$$|f(x) - f(y)| \leq K|x - y| < K\delta = \epsilon.$$

Hence f is uniformly continuous. □

Example 6.17. Let $f : [0, 2] \to \mathbb{R}$ be given by $f(x) = \sqrt{x}$. Then f is not Lipschitz, but is uniformly continuous.

6.5. **Cauchy sequence criterion and Extension Theorem.**

Theorem 6.18. *Let $f : A \to \mathbb{R}$ be a uniformly continuous function. If (x_n) is a Cauchy sequence in A, then $(f(x_n))$ is a Cauchy sequence in $\mathbb{R}$.*

Proof. Suppose (x_n) is a Cauchy sequence in A. Let $\epsilon > 0$.
By the definition of uniform continuity, there exists $\delta > 0$ such that for any $x, y \in A$ satisfying $|x - y| < \delta$,

$$|f(x) - f(y)| < \epsilon.$$

Since (x_n) is Cauchy, there exists $N \in \mathbb{N}$ such that for any $n, m > N$,

$$|x_n - x_m| < \delta.$$

Hence, combining above, for any $n, m > N$,

$$|f(x_n) - f(x_m)| < \epsilon.$$

□

Example 6.19.
(1) Let $f(x) = \sin\left(\frac{1}{x}\right)$ for $x \neq 0$. The function is not uniformly continuous on $(0, b]$ for any $b > 0$.
(2) Let $f(x) = x$ and $g(x) = \sin(x)$. Show that the product fg is not uniformly continuous.

Theorem 6.20. *Let $f : (a, b) \to \mathbb{R}$ be continuous. Then f is uniformly continuous on (a, b) if and only if the one-sided limits*

$$\lim_{x \to a^+} f(x) \quad \textit{and} \quad \lim_{x \to b^-} f(x)$$

exist.

Proof. If direction. Suppose the one-sided limits exist. Define

$$L_a = \lim_{x \to a^+} f(x), \qquad L_b = \lim_{x \to b^-} f(x).$$

Extend f to $\widetilde{f} : [a, b] \to \mathbb{R}$ by

$$\widetilde{f}(x) = \begin{cases} L_a, & x = a, \\ f(x), & x \in (a, b), \\ L_b, & x = b. \end{cases}$$

Then $\widetilde{f}$ is continuous on $[a, b]$. By Theorem 6.8, $\widetilde{f}$ is uniformly continuous on $[a, b]$. Hence its restriction f is uniformly continuous on (a, b).
Only if direction. Suppose f is uniformly continuous on (a, b). Let (u_n) be any sequence in (a, b) with $u_n \to a$. Then (u_n) is Cauchy. By Theorem 6.18, $(f(u_n))$ is Cauchy, hence convergent in $\mathbb{R}$.

Now let (x_n) be another sequence in (a, b) with $x_n \to a$. Then $|x_n - u_n| \to 0$. By uniform continuity,

$$|f(x_n) - f(u_n)| \to 0.$$

Therefore $(f(x_n))$ converges to the same limit as $(f(u_n))$. By the sequential criterion for limits, $\lim_{x\to a^+} f(x)$ exists. The proof for $\lim_{x\to b^-} f(x)$ is similar. □

Exercise 6.21. Reinterpret Example 6.19 by Theorem 6.20.

Exercise 6.22. For $b > 0$, let $g : (0, b) \to \mathbb{R}$ be given by

$$g(x) = x \sin\left(\frac{1}{x}\right).$$

Since

$$\lim_{x\to 0^+} g(x) = 0 \quad \text{and} \quad \lim_{x\to b^-} g(x) = b \sin\left(\frac{1}{b}\right),$$

the function g can be extended to a continuous function on $[0, b]$. Hence g is uniformly continuous on $(0, b)$.

6.6. **More examples.**

Exercise 6.23. Determine if the following functions are uniformly continuous. Explain briefly:

(1) $f : [0, \frac{\pi}{4}] \to \mathbb{R}$, $f(x) = \tan(x)$;
(2) $g : [\frac{\pi}{4}, \frac{\pi}{2}) \to \mathbb{R}$, $g(x) = \tan(x)$;
(3) $h : \mathbb{R} \to \mathbb{R}$, $h(x) = \frac{1}{2+\cos(x)}$.